



Biotransformation of organic nitrates by glutathione *S*-transferases and other enzymes: An appraisal of the pioneering work by William B. Jakoby

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ABSTRACT

Organic nitrates ($R-ONO_2$; R , organic residue) such as nitroglycerin are used as drugs in part for more than a century. Their pharmacological use is associated with clinically relevant tolerance which is reportedly known since 1888. The underlying mechanisms of both, the mechanisms of action and the main pharmacological effect, which is vasodilation and reduction of blood pressure, and the development of tolerance, which means increasing need of drug amount in sustained long-term therapy, are still incompletely understood. William B. Jakoby and associates were the first to report the biotransformation of organic nitrates, notably including nitroglycerin (i.e., glycerol trinitrate; GTN), by glutathione *S*-transferase (GST)-catalyzed conjugation of glutathione (GSH) to the nitrogen atom of one of the three nitrate groups of GTN to generate glutathione sulfenyl nitrite (glutathione thionitrate, *S*-nitroglutathione; GSNO₂). Jakoby's group was also the first to suggest that GSNO₂ reacts with a second GSH molecule to produce inorganic nitrite (ONO^-) and glutathione disulfide (GSSG) without the catalytic involvement of GST. This mechanism has been adopted by others to the biotransformation of GTN by mitochondrial aldehyde dehydrogenase (mtALDH-(CysSH)₂) which does not require GSH as a substrate. The main difference between these reactions is that mtALDH forms an internal thionitrate (mtALDH-(CysSH)-CysSNO₂) which releases inorganic nitrite upon intra-molecular reaction to form mtALDH disulfide (mtALDH-(CysS)₂). Subsequently, ONO^- and GSNO₂ are reduced by several proteins and enzymes to nitric oxide (NO) which is a very potent activator of soluble guanylyl cyclase to finally relax the smooth muscles thus dilating the vasculature. GSNO₂ is considered to rearrange to GSNO which undergoes further reactions including GSNO and GSSG formation. The present article is an appraisal of the pioneering work of William B. Jakoby in the area of the biotransformation of organic nitrates by GST. The two above mentioned enzymatic reactions are discussed in the context of tolerance development to organic nitrates, still a clinically relevant pharmacological concern.

1. Biotransformation and bioactivation of organic nitrates

Organic nitrates are metabolized by several enzyme families. The biotransformation of organic nitrates is outlined below for the most relevant enzymes.

1.1. Biotransformation of organic nitrates by GSH *S*-transferase

In 1950, Heppel and Hilmoe observed glutathione (GSH)¹-dependent conversion of nitroglycerin (glycerol trinitrate, GTN) and erythritol tetranitrate in liver and other tissues to inorganic nitrite (ONO^-) and glutathione disulfide (GSSG) [1]. It was hypothesized that an enzyme was responsible for these reactions. It was called “nitrite-forming

enzyme”, but had not been further characterized at that time. “Nitrite-forming activity” was found in homogenates of rabbit lung, rat kidney and rat liver [1]. GSH, but not air, was found to be essential for the conversion of GTN to nitrite and GSSG, and the GSH-to-inorganic nitrite stoichiometry was found to be about 2:1 [1]. Jakoby's group has later identified this enzyme as GSH *S*-transferase (GST) [2]. The organic nitrates nitroglycerin, erythritol tetranitrate, isosorbide dinitrate (ISDN) and ethyleneglycol dinitrate were found to be converted to nitrite and GSSG with a stoichiometry of 1:1 by rat liver GST isoforms A and B [2]. For the GST-catalyzed conversion of GTN to nitrite and GSSG, Jakoby's group was the first to formulate Eq. (1) and to suggest a mechanism for this reaction [3]. It was proposed that GSTs catalyze the nucleophilic attack of the sulfhydryl (SH) group of GSH on one of the nitro groups of GTN, which are the electrophilic centers of the molecules, to form the

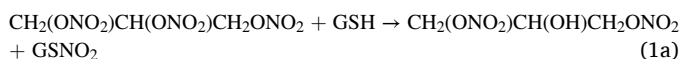
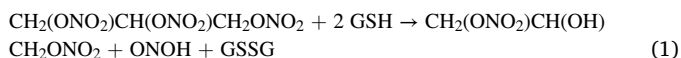
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Abbreviations used in the article

ACS	Acute coronary syndrome	GSNO ₂	S-Nitroglutathione
ALDH2	Aldehyde dehydrogenase-2	GSNO	S-Nitrosoglutathione
mtALDH	Mitochondrial aldehyde dehydrogenase	GTN	Glycerol trinitrate (nitroglycerin)
ASA	Acetylsalicylic acid	ISDN	Isosorbide dinitrate
CA	Carbonic anhydrase	ISMN	Isosorbide mononitrate
hCA	Human carbonic anhydrase	LC-MS/MS	Liquid chromatography-tandem mass spectrometry
CAD	Coronary artery disease	NO	Nitric oxide
CHF	Congestive heart failure	PE	Pentaerythritol
CVD	Cardiovascular disease	PEDN	Pentaerythritol dinitrate
GAPDH	Glyceraldehyde-3-phosphate dehydrogenase	PEMN	Pentaerythritol mononitrate
sGC	Soluble guanylyl cyclase	PETN	Pentaerythritol tetranitrate
1,2-GDN	1,2-Glyceryl dinitrate	PETriN	Pentaerythritol trinitrate
1,3-GDN	1,3-Glyceryl dinitrate	RBC	Red blood cells
1-GMN	1-Glyceryl mononitrate	RONO ₂	Organic nitrate
2-GMN	2-Glyceryl mononitrate	ROS	Reactive oxygen species
GSH	Glutathione	RSH	Organic thiol
GSSG	Glutathione disulfide	RSNO ₂	Thionitrate
cGMP	Cyclic guanosine monophosphate	SNACET	S-Nitroso-N-acetylcysteine ethyl ester
GST	Glutathione S-transferase	SNP	Sodium nitroprusside
		SOD	Superoxide dismutase
		XOR	Xanthine oxidoreductase

sulfenyl nitrite of GSH or glutathione thionitrate, i.e., S-nitroglutathione (GSNO₂), as the unstable intermediate which has not been identified thus far. It was hypothesized that GSNO₂ reacts non-enzymatically with a second GSH molecule to form GSSG and to release nitrous acid (ONOH; pK_a, 3.4) which deprotonates to form nitrite [3]. The reaction of organic nitrates (RONO₂) with thiols (RSH) to form thionitrates (RSNO₂) has been recently modelled [4]. The reaction expressed by Eq. (1b) is a redox reaction, in which the N atom with the oxidation state +5 takes up each one electron from the S (oxidation state −2) atoms of GSH and GSNO₂ to form the disulfide GSSG, with the S atoms being in the oxidation state of −1 each. The GST-catalyzed reaction of GTN to form nitrite and GSSG proposed by Jakoby's group is depicted in Scheme 1A.



Mechanistic studies by Jakoby's group with rat liver GST A, AA, B and C, and human liver GST β and δ revealed specific activities (μmol nitrite/min × mg) of 0.06, 0.14, 0.09 and 0.37, and of 0.05 and 0.32 for GTN, respectively. The corresponding data for ISDN were 0.015, 0.002, 0.004 and 0.014 and 0.019 μmol nitrite/min × mg, respectively. These data indicate that GTN is a considerably better substrate for all GSTs investigated than ISDN [3]. On the other hand, erythrityl tetranitrate was a better substrate than GTN (except for GST C). The corresponding data for erythrityl tetranitrate were 0.76, 0.28, 0.36 and 0.15, and 0.27 and 0.89 μmol nitrite/min × mg, respectively. The Michaelis-Menten constant values (K_M) were determined to be (all in mM GTN) 6.5, 0.4, 0.8 and 1.6 for GST A, AA, B and C, respectively, and 0.5 mM for GST δ [3]. The K_M values were determined to be (all in mM erythrityl tetranitrate) 0.77, 0.18, 0.23 and 0.09 for GST A, AA, B and C, respectively, and 0.07 mM for GST δ [3]. The GST-catalyzed biotransformation of GTN and ISDN to nitrite was strongly dependent upon GSH; N-acetyl-cysteine, 2-mercaptoethanol or dithiothreitol were found not to replace GSH as a substrate for GSTs underlying the high specificity of GSH as a substrate of GSTs [3]. With respect to the enzymatic mechanism it was proposed an Ordered Bi Bi reaction in which GSH is the first substrate that binds in the G-site (GSH binding site) of GST [3].

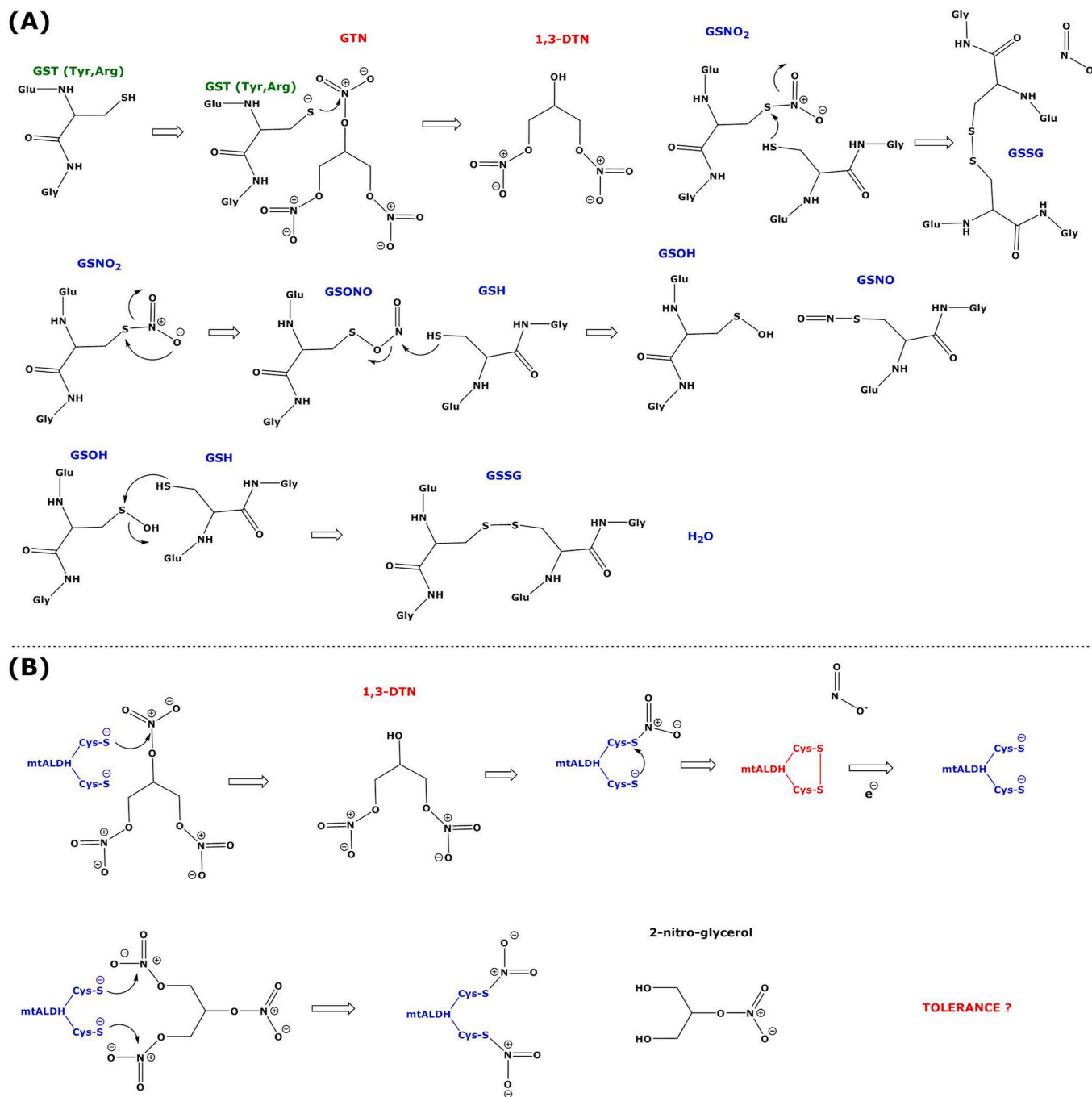
The pK_a value of the SH group of GSH was determined to be 9.2 [5]. It has been calculated that GST may decrease the pK_a value of the SH group of GSH by 2.5 units [6]. The deprotonation of the SH group of GST-bound GSH is assumed to occur via the cooperation of a catalytic tyrosine (Tyr) residue and an arginine (Arg) residue [7] (Scheme 1A). In human GSTk, Ser¹⁶ at the G-site has been proposed to function as the catalytic residue in the deprotonation of the SH group of GSH, whereas Asp⁶⁹, Ser²⁰⁰, Asp²⁰¹ and Arg²⁰² form a network of interactions with the γ-glutamyl carboxylate moiety of GSH to stabilize the thiolate anion GS[−] [8]. The H-site (the hydrophobic substrate binding site) has been proposed to be a large hydrophobic pocket with conformational flexibility to allow the binding of structurally different hydrophobic substrates.

1.2. Biotransformation of organic nitrates by glyceraldehyde-3-phosphate dehydrogenase

In the year 1975, You et al. reported that incubation of GTN with glyceraldehyde-3-phosphate dehydrogenase (GAPDH) forms inorganic nitrite in the absence of exogenous thiols [9], suggesting another enzymatic possibility for the conversion of GTN to nitrite. You et al. suggested the formation of a thioperoxide intermediate of the enzyme [9]. Yet, Keen et al. [3] suggested that in the case of GAPDH a labile protein sulfenyl nitrite (RSNO₂) can account for the products observed by You et al. [9], analogous to the GST-catalyzed GSNO₂ formation proposed by Jakoby's group. GAPDH was found to be oxidized and inactivated by GTN [9]. In vitro, GAPDH can be S-nitrosylated by low-molecular-mass S-nitrosothiols such as synthetic S-nitrosoglutathione (GSNO) on certain cysteine moieties [10].

1.3. Biotransformation of organic nitrates by aldehyde dehydrogenase

Aldehyde dehydrogenase-2 (ALDH2), i.e., mitochondrial ALDH (mtALDH), belongs to the aldehyde dehydrogenase family. ALDH2 was found to denitrate GTN to form 1,2-glyceryl dinitrate (1,2-GDN) and inorganic nitrite. ALDH2 was also observed to catalyze the reduction of GTN to NO, yet only in the presence of high amounts (1 kU/mL) of exogenous superoxide dismutase (SOD) with a very low yield of about 0.6% from 100 μM GTN in aqueous buffer [11]. This observation suggested that ALDH2 represents another enzyme being able to convert GTN to inorganic nitrite and NO (Scheme 1B) under concurrent formation of superoxide.



Scheme 1. (A) Proposed simplified mechanism for the GST-catalyzed conjugation of GST-bound GSH to GTN to form 1,3-GDN and GSNO₂ and subsequent chemical reactions to form GSSG and nitrite. (B) Proposed simplified mechanism for the mtALDH-catalyzed conversion of GTN to 1,3-GDN via S-nitrosylation of a Cys moiety of mtALDH eventually leading to oxidation of mtALDH and nitrite formation. Panel (B) proposes a double nitration of the SH groups of Cys moieties of mtALDH which may lead to tolerance development.

1.4. Organic nitrates and red blood cells

Human red blood cells (RBCs) contain considerable amounts of carbonic anhydrase (CA) I and II. The inherent catalytic activity of CA is the reversible hydration of carbon dioxide to bicarbonate. In two groups of patients with coronary artery disease (CAD) treated with GTN (intravenous injection of 5 mg/day on two days) and isosorbide dinitrate (ISDN; ingestion of 180 mg/day on three days), respectively, the activity of RBC CAI and CAII was found to be considerably inhibited [12]. In the same study, intravenous infusion of sodium nitroprusside (SNP) at a rate of 60 µg SNP/min over 10 min was also found to inhibit RBCs CAI and

CAII activity to an extent comparable to that observed for GTN and ISDN [12]. The underlying mechanism of the inhibition by GTN, ISDN and SNP of RBCs CAI and CAII activity is unclear. It has been hypothesized that GTN and ISDN are metabolized in RBCs to NO which inhibits the CA activity through S-nitrosylation of cysteine moieties in the active center of CA, thus lowering the intracellular pH value. This study suggested CA as an additional enzyme family able to interact with organic nitrates.

Human RBCs are equipped with several enzymes which can metabolize organic nitrates to S-nitrosylating species. GST ρ was first reported by Jakoby's group to be present in human RBCs (at a concentration of about 1.2 mg/100 mL packed RBC) [13]. GST ρ requires as the second

substrate GSH [13], which is present in human RBCs at concentrations of about 2 mM. However, the activity of GST ρ towards GTN and erythrityl tetranitrate was near the detection limit of the method [13]. Although it cannot be excluded, it is unlikely that GST ρ in RBCs can catalyze the conversion of organic nitrates to GSNO₂ which could form nitrite and nitrate.

1.5. Additional enzymes metabolizing organic nitrates and vascular tissue

In the presence of NADH, xanthine oxidoreductase (XOR) is able to reduce organic nitrates to nitrite and NO, as well as inorganic nitrate and inorganic nitrite to NO with very low yield, yet under strongly hypoxic conditions (<1% O₂) and at very high pharmacologically irrelevant concentrations of the organic nitrates [14]. Further enzymes and proteins reported to metabolize organic nitrates include cytochrome P450 [15], NADPH reductase and hemoglobin (for a review see Ref. [16]).

In homogenates of dog carotid artery, GTN was found to release NO and to form nitrite under anaerobic conditions [17]. NO release from GTN in vascular homogenates was found to correlate directly with the activity of GST, with GSH decreasing the release of NO [17]. It was concluded that GST is not involved in the direct bioactivation of GTN to NO, neither is NO produced from nitrite under these conditions. Hence, it was hypothesized that other at that time unknown enzymes convert GTN directly to NO [17]. Although GSTs do not need oxygen as a substrate or cofactor, these authors have not investigated GTN metabolism under aerobic conditions. A study by Chen, Zhang and Stamler has later suggested that a nitrate reductase, i.e., the mtALDH, is able to catalyze GTN biotransformation to inorganic nitrite in aerated buffer [18] (see also [19]). Yet, in that study no GSH seems to have been used [18]. The results by Kurz et al. [17] and by Chen et al. [18] are contradictory. Kurz et al. suggested direct conversion of GTN to NO [17], while Chen et al. suggested formation of nitrite from GTN [18]. Based on the study reported by J.S. Stamler's group [18], L.J. Ignarro wrote an editorial in the same journal, concluding that the molecular mechanism of action of GTN is now revealed [20]. Yet, the bioactivation of GTN and the tolerance development against organic nitrates are considered still enigmatic and not completely resolved [21,22].

GTN is known for many decades to be extensively metabolized in the liver [23]. In this work it has been for the first time suggested that GTN exerts its vasodilating effects by binding to a not yet identified receptor [23]. Furthermore, it has been suggested that tolerance to GTN may be due to altered receptor response to GTN [24]. The issue of tolerance development to organic nitrates and its clinical relevance is treated in section 3 in more detail.

1.6. Similarities and differences in the GST- and mtALDH-catalyzed biotransformation of organic nitrates

As suggested by W.B. Jakoby and associates, the mechanism of the GST-catalyzed biotransformation of GTN and other organic nitrates is assumed to include formation of GSNO₂ [2] (Scheme 1A). The mechanism of the mtALDH-catalyzed biotransformation of GTN is assumed not to involve GSH, although GSH is present in mitochondria where it is protected from drugs [25]. Moreover, it is assumed that organic nitrates such as GTN react with their nitro groups with the SH group of one of the two catalytically involved cysteine (Cys) residues of mtALDH, i.e., mtALDH-(Cys-SH)₂, to form mtALDH-(Cys-SH)-Cys-SNO₂ (Scheme 1B). mtALDH-Cys-SNO₂ and GSNO₂ from the reaction of RONO₂, GSH and GST are highly reactive compounds and are likely to undergo several reactions, including S-nitros(yl)ation reactions. Yet, S-nitrothiols and S-nitrosothiols are not NO donors by their own, but they require additional reaction partners that provide electrons, notably thiols and transition metal ions such as Cu²⁺ including their complexes as they occur in various cells and their organelles. It has been proposed that the SH group of a second Cys moiety of mtALDH reacts intra-molecularly to form the proteinic disulfide mtALDH-(Cys-S)₂ thereby releasing nitrite (Scheme

1B). Yet, it has not been considered that mtALDH-(Cys-SH)-Cys-SNO₂ could also react with mitochondrial GSH to form free GSNO₂, thereby regenerating the enzyme to form mtALDH-(Cys-SH)₂. This reaction would be equivalent to the reaction of the GST-product GSNO₂ with GSH (Scheme 1B), which is abundantly present both in the cytosol and in the mitochondria of various cells including endothelial cells [25]. Thus, both GSTs and mtALDH should be able to convert organic nitrates to nitrite via GSNO₂. In subsequent reactions, GSNO₂ is likely to undergo chemical and enzymatic reactions that eventually release NO and exert cGMP-dependent biological actions.

An important reaction of GSNO₂ is the transfer of its nitryl group (⁺NO₂) to SH groups of low-molecular-mass and high-molecular-mass thiols. These S-transnitration reactions occur very rapidly and the half-lives of relatively labile compounds such as GSNO₂ and CysSNO₂ can be extended considerably in the presence of large excess of GSH over both the organic nitrates and endogenous CysSH (Fig. 1) [26,27]. The high molar GSH:CysSH ratio also ensures the consistent low-extent formation of CysSNO which is a potent NO donor in the presence of very low concentrations of Cu¹⁺ that are formed from the reaction of ubiquitous Cu²⁺ which CysSH (Eq. (2)) [26]. Thionitrates of free GSH and CysSH and of Cys moieties in proteins and enzymes such as GST and ALDH are thought to be very labile and to not release NO [28,29]. Yet, the results of Figs. 1 and 2 suggest that thionitrates (e.g., GSNO₂, CysSNO₂) and thionitrites (e.g., GSNO, CysSNO) may be stable and, under certain conditions, may release NO (Eq. (3)).



A further important issue that also needs to be considered in this context is the role of co-factors and co-substrates, i.e., GSH for GST, and NAD⁺ for GAPDH and mtALDH. Given the relatively small amounts of ingested pharmacological organic nitrates, even in chronic use, and the effective and abundant provision of the cells with GSH and NAD⁺ by means of enzymatic regenerating systems of the cells, the consumption of GSH and NAD⁺ during the reactions should be not decisive, in contrast to the classical hypothesis of GST-associated oxidation of GSH to GSSG (see Eq. (1b)). In vitro, GTN has been reported to inactivate GST isolated and purified from rabbit liver [28]. Yet, loss of GST activity of 27% was observed at the supra-pharmacological GTN concentration of 250 μM (see next section) and only in the absence of GSH. Such a situation is most likely entirely unrealistic in vivo. In the presence of GSH at 1 mM, which is considered to be a physiological concentration, GST was reported to exert full activity [28].

A question that arises from these considerations is whether and by which mechanisms and extent may be the GST- and mtALDH-catalyzed reactions with organic nitrates responsible for tolerance development (Scheme 1).

1.7. Possible interactions of organic nitrates and their metabolites with carbonic anhydrases

Inorganic nitrite is a substrate for carbonic anhydrases (CAs). Human CAs (hCAs) may release NO from inorganic nitrite under particular conditions, such as in the presence of CysSH, presumably via formation of the potent nitrosating nitrous acid (HONO) and its anhydride (N₂O₃, ON-ONO) and CysSNO [31–33]. This newly discovered activity of CAs may be involved in the bioactivation of organic nitrates-derived inorganic nitrite. Renal CAs are also involved in the reabsorption of excreted nitrite and to a much lesser extent of nitrate [34].

Direct effects of pharmacological organic nitrates on CAs are rare and mostly limited to their effects on CA activity [12,35–37]. Thus, ISMN seems not to affect CA activity, whereas the combination of an organic nitrate with a sulfonamide, i.e., NCX250, combines potent inhibition of CA activity (K_i values: 18 nM for hCAII, 44 nM for hCAIV and 30 nM for hCA XII), NO release and activation of soluble guanylyl cyclase (sGC) in

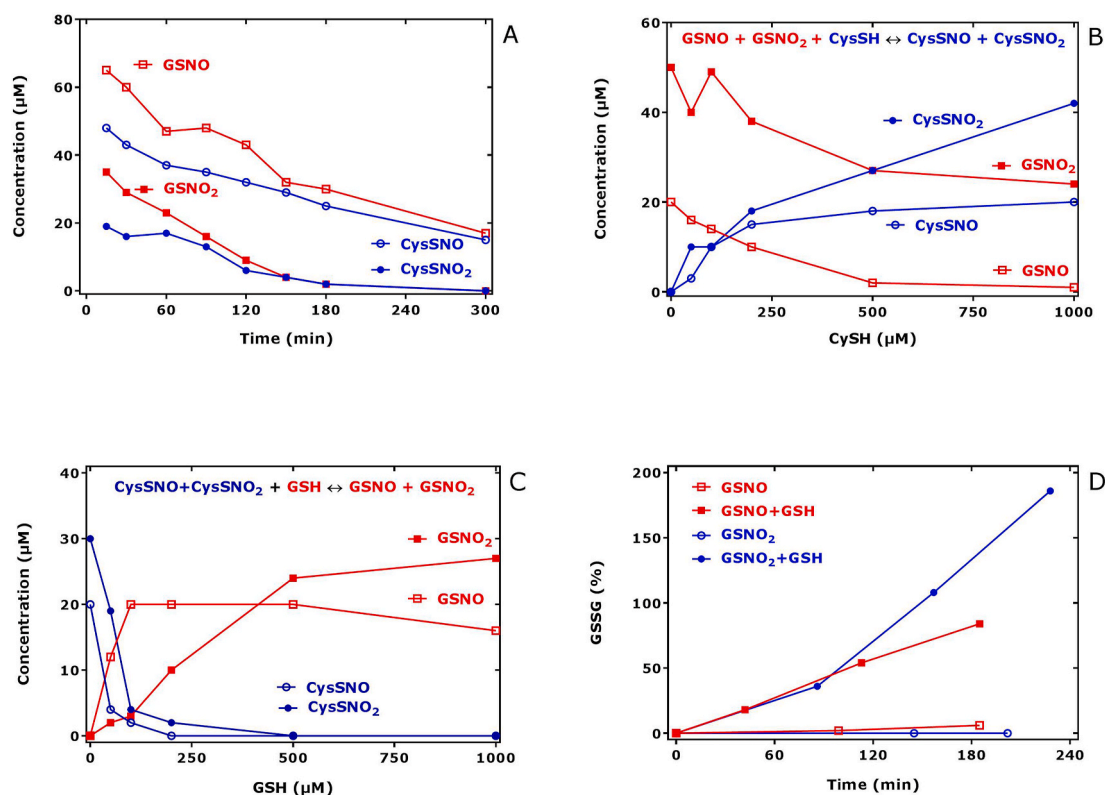


Fig. 1. Stability and S-nitrosylation reactions of S-nitroso-glutathione (GSNO), S-nitro-glutathione (GSNO₂), S-nitroso-cysteine (CysSNO), S-nitro-cysteine (CysSNO₂) in 0.2 M Tris buffer, pH 7.4, at room temperature (about 20–22 °C) in the absence and in the presence of glutathione (GSH) and cysteine (CysSH). (A) Stability of GSNO, GSNO₂, CysSNO and CysSNO₂ in the absence of GSH and CysSH. (B) Reaction of GSNO (50 μM) and GSNO₂ (20 μM) with CysSH (0–1000 μM) in the absence of GSH. The incubation time was each 2 min. (C) Reaction of CysSNO (200 μM) and CysSNO₂ (30 μM) with GSH (0–1000 μM) in the absence of CysSH. The incubation time was each 2 min. (D) Formation of glutathione disulfide (GSSG, %) from GSNO and GSNO₂ in the absence and in the presence of GSH (1 mM) for the indicated times. All analytes were measured by HPLC with UV absorbance detection at 340 nm for GSNO, GSNO₂, CysSNO, CysSNO₂ and at 205 nm for GSSG. Note: GSNO and CysSNO cannot be discriminated from GSNO₂ and CysSNO₂, respectively. The GSSG formation above 100% is presumably due to GSSG excess in the GSNO₂ preparation.

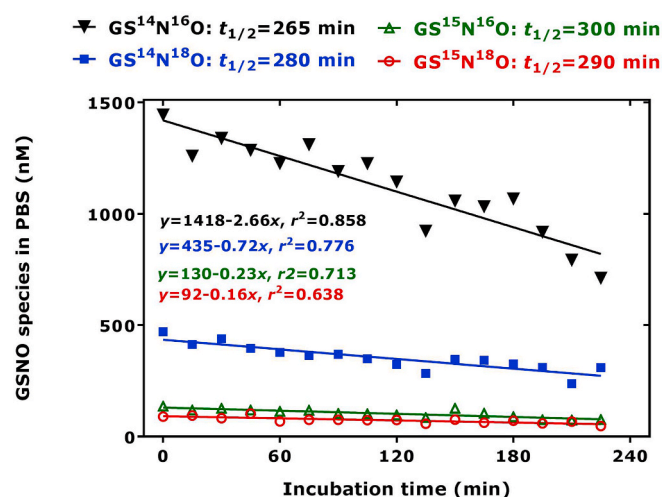


Fig. 2. Stability of S-nitrosoglutathione (GSNO) species in 67 mM potassium phosphate buffer, pH 7.4, at room temperature (about 20–22 °C). The initial nominal concentrations were 1500 nM of GS¹⁴N¹⁶O and 500 nM of GS¹⁴N¹⁸O. GSNO species were measured simultaneously by LC-MS/MS as described elsewhere [30]. GS¹⁴N¹⁸O was prepared from GSH, nitrite and 10 mM HCl in oxygen-labeled water as described previously [26]. Linear regression equations and calculated half-lives (t_{1/2}) are provided for each GSNO species.

vivo in rabbits [36]. In human erythrocytes, ISMN and GTN were found to be weak inhibitors of the hydratase activity of hCAI (K_i values, 160 μM and 220 μM, respectively) and hCAII (K_i values, 160 μM and 272 μM, respectively) [37]. In the same study, ISMN was found to be a weak inhibitor of the esterase activity of hCAI and hCAII (K_i values, 6.1 μM and 5.5 μM, respectively) [37]. The great difference in the inhibitory potency of ISMN towards the esterase and hydratase activity of CAs is interesting and remains to be explored. Whether dual drugs such as NX250, that can inhibit CA activity and concomitantly release NO from organic nitrates, are associated with tolerance development warrants further investigations.

2. Metabolism, pharmacokinetics and pharmacodynamics of organic nitrates

The clinical pharmacokinetics and pharmacodynamics of GTN, ISDN, PETN and their metabolites have been reported in detail [38–45]. In the sections that follow, the metabolism, the pharmacokinetics and the pharmacodynamics of these organic nitrates are described in brief.

2.1. Metabolism and pharmacokinetics of GTN

The clinical pharmacokinetics and pharmacodynamics of GTN, the oldest and perhaps the best investigated organic nitrate, are complex and complicated by the development of tolerance. Known GTN metabolites include 1,2-glyceryl dinitrate (1,2-GDN), 1,3-glyceryl dinitrate (1,3-GDN), 1-glyceryl mononitrate (1-GMN), 2-glyceryl mononitrate (2-GMN) and glycerol. At least five biotransformation enzymatic and non-

enzymatic pathways have been hypothesized for GTN (reviewed in Refs. [44]). Among the enzymatic pathways, the conversion of GTN to 1, 2-GDN and nitrite is considered to be responsible for GTN tolerance [44]. GTN disappears from the plasma within a few minutes, yet with prominent intra- and inter-individual variability. The plasma concentrations of the GTN metabolites exceed those of the parent molecule by several orders of magnitude. The half-lives of the metabolites are also considerably longer than that of GTN. The vasodilating effects of 1, 2-GDN and 1,3-GDN are comparable, yet these metabolites are about 10 times less potent than GTN. The role of GDN in the GTN tolerance is uncertain [44].

2.2. Metabolism and pharmacokinetics of ISDN

ISDN is metabolized to the isosorbide mononitrates (ISMN) 2-ISMN and 5-ISMN, and to isosorbide. Ingestion of a 10 mg-ISDN (42 μ mol) normal tablet by healthy subjects resulted in maximum plasma concentrations of 0.06 μ M ISDN, 0.12 μ M 2-ISMN and 0.43 μ M 5-ISMN [40].

Ingestion of a 20 mg-ISDN (85 μ mol) normal release tablet by healthy subjects resulted in maximum plasma concentrations of 0.04 μ M ISDN, 0.2 μ M 2-ISMN and 0.9 μ M 5-ISMN [41]. Ingestion of a 20 mg-ISDN (85 μ mol) sustained release tablet by healthy subjects resulted in maximum plasma concentrations of 0.03 μ M 2-ISMN and 0.18 μ M 5-ISMN, with ISDN being not detectable in plasma [41]. In the same study intravenous administration of ISDN resulted in excretion of unchanged ISDN (5 nM), 5-ISMN (1.04 μ M), 2-ISMN (0.05 μ M), and of the glucuronide of 5-ISMN (5.1 μ M) [41]. Thus, 5-ISMN and 5-ISMN glucuronide are the major ISDN metabolites in plasma and urine. Oral administration for 7 days of 120 mg 5-ISMN extended-release tablet once daily or 40 mg ISDN every 6 h resulted in maximum 5-ISMN plasma concentrations of 5.5 μ M and 3.8 μ M, respectively [45]. In these studies [38,41,45], neither inorganic nitrite nor inorganic nitrate have been measured in plasma and urine.

2.3. Metabolism and pharmacokinetics of PETN

Identified metabolites of PETN include pentaerythritol trinitrate (PETriN), pentaerythritol dinitrate (PEDN), pentaerythritol mononitrate (PEMN), and pentaerythritol (PE) [38,39]. PE and PEMN are the major PETN metabolites in urine. PETN and PETriN are not excreted in the urine. Unchanged PETN is excreted in the feces in considerable amounts [38]. Ingestion of 40 mg PETN (127 μ mol) resulted in maximum blood concentrations of 0.52 μ M PE, 0.2 μ M PEMN, and 0.02 μ M PEDN [39]. In that study, neither inorganic nitrite nor inorganic nitrate has been measured in blood and urine.

2.4. Nitrite and nitrate as metabolites of organic nitrates

Inorganic nitrate and nitrite are metabolites of all known organic nitrates and are found in blood and urine. Yet, such information is relatively scarce. As an example, oral administration of ISDN (30 mg, 127 μ mol, thrice a day) or PETN (80 mg, 253 μ mol, thrice a day) to healthy young volunteers for five days increased both the serum and the urinary concentrations of inorganic nitrite and nitrate [46]. This issue is addressed below in some detail.

ISDN ingestion increased median serum nitrate concentration from 32 μ M at baseline to 32.5 μ M at day 1 and to 38 μ M at day 5, i.e., by 6 μ M. ISDN ingestion also increased median creatinine-corrected excretion of nitrate (μ mol/mmol) from 70 at baseline to 88 at day 1 and to 98 at day 5, i.e., by 28 μ mol/mmol when compared baseline with day 5. PETN ingestion increased median serum nitrate concentration from 28 μ M at baseline to 30 μ M at day 1 and to 38 μ M at day 5, i.e., by 10 μ M. PETN treatment also increased median creatinine-corrected excretion of nitrate (μ mol/mmol) from 77 at baseline to 95 at day 1 and to 102 at day 5, i.e., by 25 μ mol/mmol when compared baseline with day 5 [46].

As to serum nitrite, both ISDN and PETN administration increased the serum nitrite concentration as shown in Fig. 3A, yet with differing

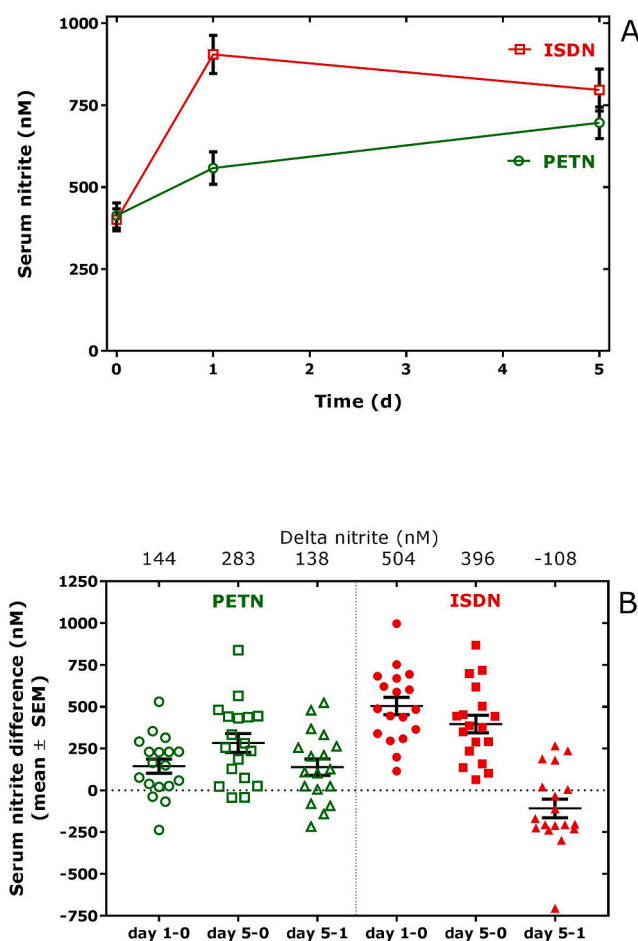


Fig. 3. Metabolism of ISDN and PETN to circulating nitrite in young healthy subjects who received the drugs for five consecutive days. (A) Serum nitrite concentration at baseline (day 0) and after 1 day and 5 days ingestion of ISDN or PETN. (B) Difference in the serum nitrite concentrations between day 1 and baseline (day 1–0), between day 5 and baseline (day 5–0), and between day 5 and day 1 (day 5–1). The Figure was constructed with data reported previously [46].

kinetics. Thus, ISDN increased mean serum nitrite concentration from 401 nM at baseline to 905 nM at day 1 and to 797 nM at day 5. PETN increased mean serum nitrite concentration from 414 nM at baseline to 558 nM at day 1 and to 696 nM at day 5. The differences in serum nitrite concentrations between the treatment days are illustrated in Fig. 3B. It is notable the large decrease in the serum nitrite concentration differences between day 5 and day 1, and between day 5 and day 0 in the case of ISDN compared to PETN (504 nM vs. 145 nM). This observation may lead to the conclusion that ISDN is stronger pronounced to “nitrate tolerance” than PETN. ISDN (on day 1), but not PETN lowered systemic blood pressure [46]. In this study, neither ISDN nor PETN induced systemic oxidative and nitritative stress. ISDN led to the stimulation of sGC on the basis of increased cGMP excretion, with PETN failing such statistical significance [46].

2.5. Pharmacodynamics and tolerance development to organic nitrates and inorganic nitrite

Inorganic nitrite and inorganic nitrate have been reported not to induce tolerance in primates [47], and to augment tolerance to ischemia/reperfusion injury [48]. NO can be produced through the $\text{NO}_3^-/\text{NO}_2^-/\text{NO}$ axis. Loss of circulating nitrite derived from organic nitrates due to impaired or inhibited activity of renal CAs, i.e., due to elevated nitrite excretion, of the drugs themselves, and of their organic

metabolites or by other mechanisms may finally diminish NO-related biological activity, thus simulating tolerance development. We proposed that the molar ratio of urinary nitrate to urinary nitrite, i.e., U_{NO_3R} , may be a useful measure of nitrite-dependent renal CA activity and relative loss of circulating nitrite: The higher the U_{NO_3R} value, the lower the nitrite loss in the urine [34]. At baseline, the U_{NO_3R} values did not differ statistically significantly between ISDN and PETN: 1570 vs 1305 ($P = 0.67$). The U_{NO_3R} values decreased remarkably for both drugs after 1 day (800 vs 480, $P = 0.04$) and after 5 days (600 vs 380, $P = 0.002$). On day 0 and day 1, both U_{NO_3R} and the area under the curve values were statistically significantly lower for ISDN compared to PETN, suggesting that use of ISDN is associated with a higher loss of NO equivalents. These observations are in line with the serum nitrite concentrations (see Fig. 3) and may translate into a higher potential of ISDN to induce tolerance compared to PETN [34].

In patients with proven CAD, stable angina pectoris and elevated pulmonary artery pressure during bicycle exercise, the acute and chronic (4 and 8 months) effects of PETN (50–300 mg) or ISDN (40–120 mg) were evaluated in comparison to sublingual GTN [49]. GTN was by about 30%–40% more effective than PETN and ISDN with regard to pulmonary artery pressure at exercise. Beneficial effects of PETN and ISDN were seen with regard to the number of angina attacks, GTN consumption, and ST-segment depression. PETN and ISDN decreased exercise pulmonary artery pressure by 15%–20%, at acute testing and during chronic therapy. There was no difference with respect to the long-term effects of PETN and ISDN, and there were no signs of nitrate tolerance development. Yet, more side-effects were observed during ISDN treatment [49].

Whether PETN is the “better” organic nitrate in pharmacotherapy with respect to the anti-anginal efficacy and tolerance development is still an important issue of research [50–56]. However, despite some evidence suggestive of possible superiority of PETN over other organic nitrates in terms of the influence on endothelial function [54], tolerance development and lipid peroxidation [55], no significant effects of PETN were demonstrated in placebo-controlled randomized studies with regard to either endothelial function [56] or exercise tolerance [51] in patients with stable CAD participating in the PENTA [56] and CLEOPATRA [51] trials, respectively. Admittedly, some beneficial microcirculatory changes were observed in the PENTA study. Nevertheless, clinical benefits in the CLEOPATRA study were limited to 120 out of 655 trial participants, i.e., to those who were symptomatic at low exercise levels [51].

3. Organic nitrates tolerance from a clinical point of view and potential roles of biotransformation pathways

The phenomenon of organic nitrate tolerance has been an intriguing issue since the discovery of acute anti-anginal properties of GTN almost 150 years ago. As early as 1888, for the first time a tolerance to GTN was described by D.D. Stewart, being referred to as a need for about 20-fold higher dosage of GTN to maintain its comparable ability to alleviate angina symptoms [57].

From the clinical point of view, nitrate tolerance limits the use of organic nitrates not only in ischemic heart diseases but also in heart failure. Numerous beneficial anti-ischemic and hemodynamic effects wane over time after only a few days of continuous therapy with organic nitrates. Nitrate tolerance is ascribed to neurohormonal activation (including sympathetic activity, renin-angiotensin-aldosterone axis stimulation and vasopressin release) and volume expansion (i.e., pseudotolerance) as well as direct vascular effects [18,50,58]. The latter are linked to an excessive generation of reactive oxygen species (ROS) in endothelial and vascular smooth muscle cells and impaired mitochondrial bioactivation of nitrates, in part owing to the ROS-induced inhibition of mtALDH, of which both activity and expression are regulated in a redox-sensitive manner. Several putative ROS sources were proposed as potential contributors, e.g., mitochondrial respiratory chain,

membrane-bound NAD(P)H oxidases and “uncoupled” NO synthase [59, 60].

Despite pivotal mechanistic studies about 20 years ago elucidating pathways responsible for nitrate tolerance [18,58], recent research did not result in the translation of these landmark discoveries from bench to bedside. According to the Scopus database (accessed on Aug 26th, 2020) during last 25 years the number of records with “nitrate tolerance” in abstract, title or keywords was highest in 1998 (50) and 2002 (37), while respective figures fell later to 17 in 2013 and even to as low as 8 in 2019. Of note, a rapid gradual loss of clinical efficacy of organic nitrates is still a well-recognized observation even in patients receiving a recommended medical therapy including drugs which reportedly attenuated nitrate tolerance in experimental studies. In particular, the putative capability to counteract nitrate tolerance was shown for angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, carvedilol, hydralazine and statins [52]. Not only have all these therapeutic strategies been shown to possess antioxidant properties, but they also constitute a standard of care in subjects with arterial hypertension, CAD and/or heart failure. Unfortunately, their ability to alleviate clinical nitrate tolerance has not been unequivocally proven and also experimental studies brought inconsistent and inconclusive results thus far.

The only practical approach to prevent clinical nitrate tolerance is a nitrate-free interval; yet, this can be potentially dangerous to patients due to rebound ischemia. The phenomenon of rebound ischemia is attributable to an enhanced sensitivity to vasoconstrictors and impaired ability of endothelium-dependent agonists to elicit vasodilation, which results in endothelial dysfunction, sometimes referred to as cross-tolerance. These abnormalities manifest as an increased frequency of angina episodes on nitrate-withdrawal. Accordingly, the use of organic nitrates may not only be of no clinical benefit (like in GISSI-3 and ISIS-4 trials in acute myocardial infarction), but they might even prove detrimental. Indeed, some observational large-scale studies revealed a higher mortality risk associated with chronic usage of long-acting organic nitrates in post-infarction patients [61]. That these detrimental effects are exclusively due to a misbalance between vasoconstrictory and vasodilatory pathways seems rather unlikely. Because endothelial dysfunction is considered an antecedent of acute coronary syndromes (ACS), endothelial activation and plaque destabilization might also occur as a consequence of impaired bioavailability of endogenous NO in organic nitrate-tolerant patients.

As to the anti-aggregatory effects of organic nitrates, it is still unclear whether they are sustained or impaired upon development of vascular tolerance [62,63]. This is of particular importance in patients with advanced atherosclerotic cardiovascular disease (CVD) who are prone to platelet aggregation on the surface of ruptured coronary plaques, which is a basis of ACS. Assuming excessive ROS generation in the vasculature of nitrate-tolerant subjects with CVD, alternative routes of the delivery of NO, an anti-aggregatory mediator, can be more efficient in order to “by-pass” the pro-oxidant and NO-scavenging milieu of the vascular wall. NO can be generated from inorganic nitrite and inorganic nitrate after its reduction to nitrite, but until recently the role of this nitrate/-nitrite/NO cycle has been underestimated at least under physiological pH in the blood and normoxic conditions. However, it was recently demonstrated that intra-platelet cyclic GMP formation may be induced upon incubation of inorganic nitrite with isolated erythrocytic CAs and human platelets [32]. The administration of inorganic nitrite and lipophilic *S*-nitrosothiols like SNACET offer attractive perspectives for future studies [64,65].

It is noteworthy that, beyond vascular effects, exposure to organic nitrates is able to protect against ischemia/reperfusion injury, thus mimicking another intriguing phenomenon, i.e., ischemic preconditioning. A preconditioning-like effect was hypothesized as a possible explanation of a marked shift in ACS presentation away from STEMI in favor of NSTEMI/ACS associated with chronic nitrate therapy before ACS onset in the Global Registry of Acute Coronary Events

(GRACE) encompassing 52,693 subjects [66]. Moreover, organic nitrate use was strongly correlated with a lower release of cardiac necrosis biomarkers in the GRACE registry.

Intriguingly, various organic nitrates trigger different mechanisms of the preconditioning-like effects. Additionally, the development of tolerance to organic nitrates-mediated preconditioning is not uniform. In a human *in vivo* study a short-term exposure to GTN or PETN prevented endothelial dysfunction which was induced by ischemia and reperfusion 24 h later. This delayed preconditioning-like protective effect could be blocked by supplementing ascorbic acid (vitamin C) only for GTN. This suggests its dependence upon GTN-dependent ROS formation [67]. However, upon prolonged GTN exposure the ability of GTN to evoke endothelial protection was lost but could be restored by vitamin C [68]. In contrast, PETN-induced preconditioning was maintained after prolonged PETN administration [69]. As mentioned above, PETN offers improved pharmacological profile than GTN and ISDN with respect to tolerance development and endothelial dysfunction, presumably via up-regulation of heme oxygenase-1 (HO-1), an enzyme involved in anti-oxidant defense [70].

With respect to GTN and ISDN many agents have been reported to prevent or avoid organic nitrate tolerance in healthy humans and in subjects suffering from CAD, congestive heart failure (CHF), or hypertension. These drugs include atorvastatin, carvedilol, folic acid, hydralazine, L-arginine, lipoic acid, N-acetylcysteine, and vitamin C against GTN; captopril, hydrochlorothiazide + amiloride, losartan, and vitamin C against ISDN and 5-ISMN. Although many of these agents are categorized as anti-oxidants, they also exert effects that cannot be simply classified as being purely anti-oxidant, but are involved in many different pathways. The chemical and biochemical diversity of agents being able to prevent or avoid tolerance development to organic nitrates mirror the complexity of their metabolism and biological actions which may only in part be related to oxidative stress. The formation of GSSG and other ROS products *in vitro* experiments with organic nitrates is likely to have led to the so-called thiol depletion theory of nitrate tolerance [71–73]. Yet, there is increasing evidence that this holds not true as discussed above (see also Ref. [74]).

Chronic use of pharmacological organic nitrates generates both beneficial and detrimental effects. The net effect is likely to dependent on a given clinical condition. As an example, Redfield et al. [75] failed to demonstrate any improvement or impairment in some indirect measures of exercise tolerance on 6-week ISMN compared to placebo in heart failure with preserved ejection fraction. Further experimental and especially clinical investigations of the mechanisms by which pharmacological organic nitrates exert activity and develop tolerance are required [76]. Based on previous achievements, but also on the insufficiency of apparently promising strategies to prevent clinical nitrate tolerance, improved methodologies are required to solve the still enigmatic life of currently used organic nitrates.

4. Conclusions and perspectives

GTN, an organic nitrate, is used as a drug since more than 130 years. Subsequently, the structurally closely related ISDN and PETN were developed and are currently clinically used. Generally, it is believed that GTN, ISDN and PETN are finally biotransformed to the potent vasodilator nitric oxide (NO). W.B. Jakoby was the first to propose that GTN and other organic nitrates are substrates of GSH S-transferases (GSTs) to produce S-nitroglutathione (GSNO₂), which is subsequently converted to inorganic nitrite and GSSG by a non-enzymatic reaction. Jakoby's central idea has been that GSTs catalyze the nucleophilic attack of the highly specific substrate glutathione (GSH) to virtually every compound that contains an electrophilic center in its molecule by increasing the concentration of the nucleophile glutathione (GS[−]) by two orders of magnitude. In the case of GTN and other organic nitrates, i.e., nitric esters of alcohols (RONO₂), the electrophilic center is the N atom of the nitric ester. The GST-catalyzed reaction of the enzyme-bound GS[−] with

RONO₂ is not a redox reaction, but is simply a classical nucleophilic substitution reaction. The formation of GSSG in this reaction in *in vitro* settings has most likely led to the general belief that the biotransformation of GTN is generally associated with oxidative stress. Yet, this is most likely not true with respect to the biotransformation of organic nitrates by GSTs. Organic nitrates are substrates for other proteins and enzymes, notably for mtALDH of which the first reaction step with RONO₂ is the formation of mtALDH thionitrate (mtALDH-CysSNO₂), analogous to the mechanism suggested by W.B. Jakoby for the GST-catalyzed biotransformation of organic nitrates to GSNO₂. Direct evidence of GST-catalyzed formation of GSNO₂ from pharmacological organic nitrates is a challenging task and has not been demonstrated thus far. Such a demonstration would be the definite proof of Jakoby's proposal.

The mechanisms of the vasodilatory effects of and tolerance development to organic nitrates, in first line to GTN, are still incompletely elucidated. We speculate that the bioactivation of organic nitrates by GSTs does not contribute to tolerance development to organic nitrates. Other redox-based proteins and enzymes, being involved in the biotransformation of organic nitrates, and ubiquitous CAs, being involved in the renal resorption of the NO-reservoir nitrite, and possibly being involved in the bioactivation of inorganic nitrite to NO and S-nitrosothiols, are more likely to contribute to clinical tolerance development to organic nitrates. Today, we have the knowledge and the analytical tools to conduct this scientific undertaking.

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